SKILLEDTRADES BC

PROGRAM OUTLINE

Asphalt Paving/Laydown Technician

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ASPHALT PAVING/LAYDOWN TECHNICIAN PROGRAM OUTLINE

APPROVED

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Developed by
SkilledTradesBC
Province of British Columbia

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Section 1

INTRODUCTION

ASPHALT PAVING/LAYDOWN TECHNICIAN

Foreword

A Program Outline is a SkilledTradesBC Program Standards communication tool. It reflects the full scope of knowledge and skills required to competently perform an occupation anywhere in B.C.

The Program Outline must guide development of curriculum and learning resources because all SkilledTradesBC assessment tools are designed to measure achievement of the competencies and learning tasks it describes for an occupation (e.g., level exams, certification exams, practical assessments).

The Program Outline informs industry, training providers, instructors, the public, apprentices, and sponsors of the occupation's requirements for certification, including:

- The program Credentialing Model
- General Areas of Competence (GACs) and specific competencies required by individuals to perform proficiently in this occupation
- Learning tasks and content that must be mastered in order for an individual to be deemed competent
- Achievement Criteria for demonstrating practical competencies

It further informs technical training delivery regarding:

- Levels at which competence mastery is required
- Suggested time allocation for each topic
- Facility requirements
- Required tools and equipment
- Reference materials
- Instructor qualifications
- Assessment guidelines

Practical demonstration and student participation should always be integrated with classroom sessions. The training program features practical demonstrations of the following:

- Raker
- Asphalt Paver
- Compaction Roller
- Screed
- Milling Machine

Safe working practices, though not always specified in each of the competencies and learning tasks, are an implied part of the program and should be stressed throughout the training.

Acknowledgements

The Program Outline was prepared with the advice and direction of an industry steering committee convened initially by the BC Road Builders and Heavy Construction Association.

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Facilitators:

- Kent Orrock, BCRBHCA
- Lee Middleton, Fulford Harbour Consulting Ltd.
- Bob McConkey, Douglas College
- Dan Wood, Stonecoast Group/Douglas College
- Miriam Mari, Stonecoast Group/Douglas College

SkilledTradesBC would like to acknowledge the dedication and hard work of all the industry representatives appointed to identify the training requirements of the Asphalt Paving/Laydown Technician occupation.

How to Use this Document

This Program Outline has been developed for the use of individuals from several different audiences. The table below describes how each section can be used by each intended audience.

Section	Training Providers	Employers/ Sponsors	Apprentices	Challengers
Program Credentialing Model	Level 1 Technical Training consists of 120 hours or 4 weeks. Generally the course is 2 weeks of theory followed by 2 weeks of practical. During the practical the apprentice is required to learn basic skills for all five endorsable positions of the asphalt team.	Understand the length and structure of the program. Work-Based Training requires a minimum of 100% of all the competencies listed for any one endorsement position (i.e., Asphalt Paver, Screed, etc.). Work-Based Training also requires the minimum hours listed for the endorsement. These are intended to be the minimum hours required before full competence is considered. Many apprentices may require additional hours before achieving full competence.	Level 1 Technical Training consists of 120 hours or 4 weeks. Generally the course is 2 weeks of theory followed by 2 weeks of practical. Upon completion of the In-School Technical Training the apprentice is required to work within the industry, gaining hours and competencies for any of the endorsements (i.e., Raker, Compaction Roller, etc.). The apprentice may be required to achieve additional hours before being considered fully competent for an endorsement. This is due to the fact that not all apprentices will progress at the same pace.	To challenge the Asphalt Paving/Laydown Technician Program an individual must achieve 150% of the required apprentice able hours plus all required practical competencies for the endorsement being challenged.
Program Assessment	Practical Assessment is completed using the Practical Assessment Form Located in Appendix C of the Asphalt Paving/Laydown Technician Program Instructors Guide. An overall percentage grade must be assessed to each apprentice and sent to the SkilledTradesBC.	Understand the various assessment requirements for the program An optional logbook has been developed and is available through Crown Publications BC. It lists and helps track both hours and required competence for each endorsement.	In-School Assessment of a minimum of 70%. Certificate of Qualification – SkilledTradesBC invigilated Exam – Minimum 70% An optional logbook has been developed and is available through Crown Publications BC. It lists and helps track both hours and required competence for each endorsement.	150% Documented hours and competencies listed for the endorsement(s) being challenged. Certificate of Qualification – SkilledTradesBC invigilated Exam – Minimum 70%

Section	Training Providers	Employers/ Sponsors	Apprentices	Challengers
OAC	Communicate the competencies that industry has defined as representing the scope of the occupation (see OAC)	Understand the competencies that an apprentice is expected to demonstrate in order to achieve certification	View the competencies they will achieve as a result of program completion	Understand the competencies they must demonstrate in order to challenge the program
Training Topics and Suggested Time Allocation	Shows proportionate representation of general areas of competency (GACs) at each program level, the suggested proportion of time spent on each GAC, and percentage of time spent on theory versus practical application	Understand the scope of competencies covered in the technical training, the suggested proportion of time spent on each GAC, and the percentage of that time spent on theory versus practical application	Understand the scope of competencies covered in the technical training, the suggested proportion of time spent on each GAC, and the percentage of that time spent on theory versus practical application	Understand the relative weightings of various competencies of the occupation on which assessment is based
Program Content	Defines the objectives, learning tasks, high level content that must be covered for each competency, as well as defining observable, measureable achievement criteria for objectives with a practical component	Identifies detailed program content and performance expectations for competencies with a practical component; may be used as a checklist prior to signing a recommendation for certification (RFC) for an apprentice	Provides detailed information on program content and performance expectations for demonstrating competency	Allows individual to check program content areas against their own knowledge and performance expectations against their own skill levels
Training Provider Standards	Defines the facility requirements, tools and equipment, reference materials (if any) and instructor requirements for the program	Identifies the tools and equipment an apprentice is expected to have access to; which are supplied by the training provider and which the student is expected to own	Provides information on the training facility, tools and equipment provided by the school and the student, reference materials they may be expected to acquire, and minimum qualification levels of program instructors	Identifies the tools and equipment a tradesperson is expected to be competent in using or operating; which may be used or provided in a practical assessment

Section 2

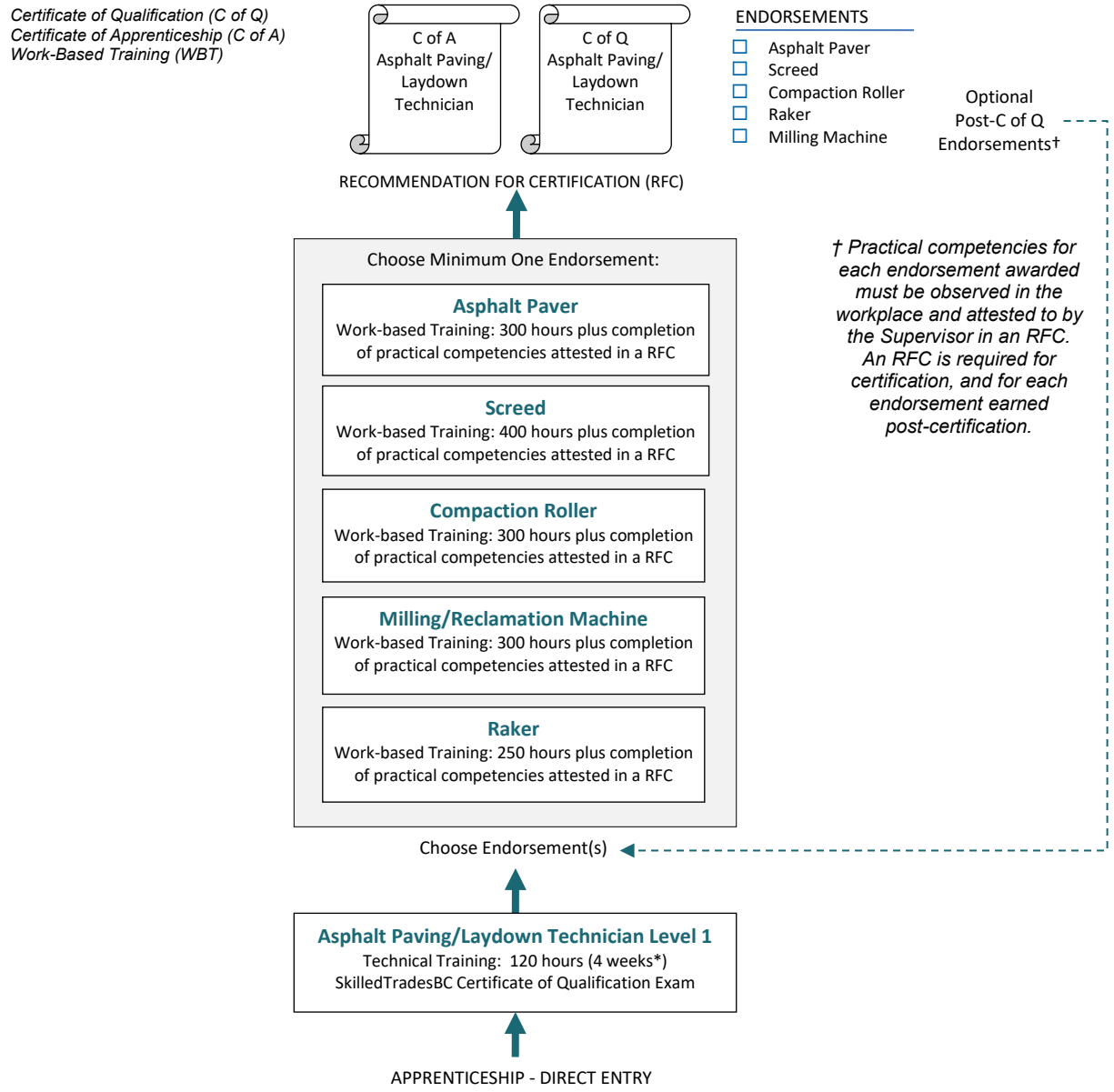
PROGRAM OVERVIEW

Asphalt Paving/Laydown Technician

Program Credentialing Model

Apprenticeship Pathway

This graphic provides an overview of the Asphalt Paving/Laydown Technician apprenticeship pathway.



**Suggested duration based on 30-hour week*

CROSS-PROGRAM CREDITS

Individuals who hold the credentials listed below are entitled to receive partial credit toward the completion requirements of this program

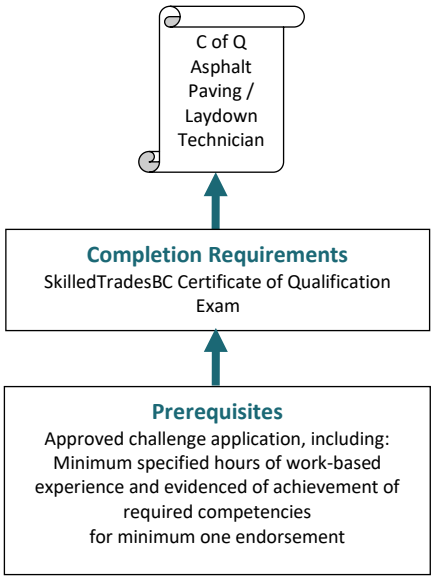
None

Program Credentialing Model

Challenge Pathway

This graphic provides an overview of the Asphalt Paving/Laydown Technician challenge pathway.

Certificate of Qualification (C of Q)



CREDIT FOR PRIOR LEARNING

Individuals who hold the credentials listed below are entitled to receive partial credit toward the completion requirements of this program

None

Program Assessment

Apprentices will be assessed fairly and accurately throughout the program on the various skills required to be a professional tradesperson. Assessment activities are designed to provide feedback and allow for further development of skills that have been identified as essential for on-the-job performance.

The forms of assessment used in this program as described below.

Completion Requirement	Evidence of Achievement	Level of Achievement Required
Level 1 Technical Training	In school testing and practical assessment	Minimum 70%
Work-based training hours	Work-based training report completed by sponsor or employer	Hours plus competencies for any one endorsement. Endorsements include: **Asphalt Paver**: 300 hours plus completion of practical competencies **Screed**: 400 hours plus completion of practical competencies **Compaction Roller**: 300 hours plus completion of practical competencies **Raker**: 250 hours plus completion of practical competencies **Milling Machine**: 300 hours plus completion of practical competencies
Certificate of Qualification Exam	SkilledTradesBC-administered exam	Minimum 70%
Recommendation for Certification	Approval or sign-off by Sponsor, Employer, or other individual with sign-off authority	Declared Competent

Occupational Analysis Chart

ASPHALT PAVING/LAYDOWN TECHNICIAN

Occupation Description: Asphalt Paving/Laydown Technicians operate machines that lay, screed, compact or mill surface materials in highway and road construction. They may also operate trucks equipped with road sanding, road oiling and other similar apparatus.

ROAD BUILDING PRINCIPLES AND INTRODUCTION TO PAVING **A**	Introduction to the paving industry and paving equipment A1	Describe history of asphalt pavement A2	Describe paving technology A3	Describe industry expectations of/for workers A4	Describe basic road and highway profiles A5	Describe future paving/ construction trends A6
ASPHALT PAVING SAFETY **B**	Safety orientation B1	Describe safety working around haul trucks B2	Describe safety working around and operating power brooms B3	Describe safety working around and operating air hammers B4	Describe safety working around and operating propane tanks B5	Describe weather effects and precautions B6
	Describe safety and traffic control procedures B7	Describe plant site and gravel pit safety B8	Describe light vehicle transporting procedures B9	Complete fatigue awareness training B10		
ASPHALT PRODUCTION AND PRODUCTS **C**	Describe crushing process and paving aggregate production C1	Describe types and specifications of aggregates C2	Describe asphalt plant operation C3	Describe asphalt products and uses C4		
ASPHALT ENVIRONMENTAL CONSIDERATIONS **D**	Describe environmentally friendly properties D1	Describe use of storage areas D2	Describe asphalt plant and placement emissions D3			
ASPHALT TESTS AND QUALITY CONTROL **E**	Determine asphalt mix design E1	Confirm that samples match mix design E2	Determine haul truck samples E3	Determine compaction levels E4		

ROAD STRUCTURES — F

- Describe drainage, grade and base — F1
- Verify slopes, crowns and grades — F2
- Describe final grade preparation — F3

END PRODUCT SPECIFICATIONS — G

- Describe gradation — G1
- Describe oil content — G2
- Describe application — G3
- Describe segregation — G4
- Describe density — G5
- Describe smoothness — G6

OPERATE MILLING AND RECLAIMING EQUIPMENT — H

- Describe milling and reclaiming equipment — H1
- Identify milling and reclaiming safety procedures — H2
- Describe milling and reclaiming operator roles and responsibilities — H3
- Perform milling and reclaiming equipment maintenance — H4
- Comply with markers, grades and stakes — H5
- Operate milling and reclaiming equipment — H6
- Follow shut down procedures — H7
- Transport milling and reclaiming equipment — H8

ASPHALT PAVER OPERATION — I

- Describe asphalt placement process — I1
- Describe paving machine equipment and attachments — I2
- Apply paver work safety procedures — I3
- Describe asphalt paver operator roles and responsibilities — I4
- Perform paving machine maintenance — I5
- Comply with markers, grades and stakes — I6
- Operate asphalt paver — I7
- Follow shut down procedures — I8
- Transport paver — I9

ASPHALT SCREED OPERATION — J

- Describe paving screed and attachments — J1
- Apply screed work safety procedures — J2
- Describe asphalt screed operator roles and responsibilities — J3
- Perform screed maintenance — J4
- Operate asphalt screed — J5
- Follow shut down procedures — J6

RAKING OPERATION **K**	Describe raker roles and responsibilities K1	Apply raking safety procedures K2	Place asphalt hot mix K3
COMPACTOR OPERATION **L**	Describe compaction equipment L1	Apply compaction work safety procedures L2	Describe asphalt Compaction Roller operator roles and responsibilities L3

Perform Compaction Roller maintenance L4	Operate Compaction Roller L5	Follow shut down procedures L6

Training Topics and Suggested Time Allocation

ASPHALT PAVING/LAYDOWN TECHNICIAN

			% of Time Allocated to:		
		% of Time	Theory	Practical	Total
Line A	**ROAD BUILDING PRINCIPLES AND INTRODUCTION TO PAVING**	**3%**	**100%**	**0%**	**100%**
A1	Introduction to the paving industry and paving equipment		✓		
A2	Describe history of asphalt pavement		✓		
A3	Describe paving technology		✓		
A4	Describe industry expectations of/for workers		✓		
A5	Describe basic road and highway profiles		✓		
A6	Describe future paving/ construction trends		✓		
Line B	**ASPHALT PAVING SAFETY**	**9%**	**90%**	**10%**	**100%**
B1	Safety orientation		✓		
B2	Describe safety working around haul trucks		✓	✓	
B3	Describe safety working around and operating power brooms		✓	✓	
B4	Describe safety working around and operating air hammers		✓		
B5	Describe safety working around and operating propane tanks		✓	✓	
B6	Describe weather effects and precautions		✓		
B7	Describe safety and traffic control procedures		✓		
B8	Describe plant site and gravel pit safety		✓		
B9	Describe light vehicle transporting procedures		✓		
B10	Complete fatigue awareness training		✓		
Line C	**ASPHALT PRODUCTION AND PRODUCTS**	**7%**	**100%**	**0%**	**100%**
C1	Describe crushing process and paving aggregate production		✓		
C2	Describe types and specifications of aggregates		✓		
C3	Describe asphalt plant operation		✓		
C4	Describe asphalt products and uses		✓		
Line D	**ASPHALT ENVIRONMENTAL CONSIDERATIONS**	**5%**	**100%**	**0%**	**100%**
D1	Describe environmentally friendly properties		✓		
D2	Describe use of storage areas		✓		
D3	Describe asphalt plant and placement emissions		✓		
Line E	**ASPHALT TESTS AND QUALITY CONTROL**	**6%**	**80%**	**20%**	**100%**
E1	Determine asphalt mix design		✓		
E2	Confirm that samples match mix design		✓		
E3	Determine haul truck samples		✓	✓	
E4	Determine compaction levels		✓		

			% of Time Allocated to:		
		% of Time	Theory	Practical	Total
Line F	**ROAD STRUCTURES**	**6%**	**60%**	**40%**	**100%**
F1	Describe drainage, grade and base		✓		
F2	Verify slopes, crowns and grades		✓	✓	
F3	Describe final grade preparation		✓		
Line G	**END PRODUCT SPECIFICATIONS**	**8%**	**100%**	**0%**	**100%**
G1	Describe gradation		✓		
G2	Describe oil content		✓		
G3	Describe application		✓		
G4	Describe segregation		✓		
G5	Describe density		✓		
G6	Describe smoothness		✓		
Line H	**OPERATE MILLING AND RECLAIMING EQUIPMENT**	**10%**	**20%**	**80%**	**100%**
H1	Describe milling and reclaiming equipment		✓		
H2	Identify milling and reclaiming safety procedures		✓		
H3	Describe milling and reclaiming operator roles and responsibilities		✓		
H4	Perform milling and reclaiming equipment maintenance		✓	✓	
H5	Comply with markers, grades and stakes		✓	✓	
H6	Operate milling and reclaiming equipment		✓	✓	
H7	Follow shut down procedures		✓	✓	
H8	Transport milling and reclaiming equipment		✓	✓	
Line I	**ASPHALT PAVER OPERATION**	**10%**	**20%**	**80%**	**100%**
I1	Describe asphalt placement process		✓		
I2	Describe paving machine equipment and attachments		✓		
I3	Apply paver work safety procedures		✓	✓	
I4	Describe asphalt paver operator roles and responsibilities		✓		
I5	Perform paving machine maintenance		✓	✓	
I6	Comply with markers, grades and stakes		✓	✓	
I7	Operate asphalt paver			✓	
I8	Follow shut down procedures		✓	✓	
I9	Transport paver		✓	✓	
Line J	**ASPHALT SCREED OPERATION**	**10%**	**20%**	**80%**	**100%**
J1	Describe paving screed and attachments		✓		
J2	Apply screed work safety procedures		✓	✓	
J3	Describe asphalt screed operator roles and responsibilities		✓		
J4	Perform screed maintenance		✓	✓	
J5	Operate asphalt screed		✓	✓	
J6	Follow shut down procedures		✓	✓	
Line K	**RAKING OPERATION**	**10%**	**20%**	**80%**	**100%**
K1	Describe raker roles and responsibilities		✓		
K2	Apply raking safety procedures		✓	✓	

			% of Time Allocated to:		
		% of Time	Theory	Practical	Total
K3	Place asphalt hot mix		✓	✓	
Line L	**COMPACTOR OPERATION**	**16%**	**20%**	**80%**	**100%**
L1	Describe compaction equipment		✓		
L2	Apply compaction work safety procedures		✓	✓	
L3	Describe asphalt Compaction Roller operator roles and responsibilities		✓		
L4	Perform Compaction Roller maintenance		✓	✓	
L5	Operate Compaction Roller		✓	✓	
L6	Follow shut down procedures		✓	✓	
	Total Percentage for Asphalt Paving/Laydown Technician	**100%**			

Section 3

PROGRAM CONTENT

Asphalt Paving/Laydown Technician

Level 1

Asphalt Paving/Laydown Technician

Line (GAC): A ROAD BUILDING PRINCIPLES AND INTRODUCTION TO PAVING

Competency: A1 Introduction to the paving industry and paving equipment

Objectives

To be competent in this area, the individual must be able to understand roads construction history and the relationship between governing bodies and the contractors.

LEARNING TASKS	CONTENT
1. Describe paving and what types of equipment are used	• Basic process from pit to finished product
2. Describe how roads were originally created	• Basic overview of history
3. Describe the importance of roads	• Infrastructure
4. Describe provincial road paving resources	• Gravel, aggregates
5. Explain who uses paving	• Recreation, trade, we all rely on it
6. Demonstrate understanding of relevant statistics	• Size of industry • Economics • Number of employers
7. Describe other types of roads	• Ice roads, etc.
8. Review relevant regulations and funding	• Government regulations like EPS, MMCD, etc.
9. Discuss environmentally friendly	• Asphalt vs. concrete

Line (GAC): A ROAD BUILDING PRINCIPLES AND INTRODUCTION TO PAVING

Competency: A2 Describe history of asphalt pavement

Objectives

To be competent in this area, the individual must be able to understand the progress and evolution of road construction techniques.

LEARNING TASKS	CONTENT
1. Describe evolution of roads	• Ancient roads e.g., Appian Way and their evolution
2. Describe products used	• Original paving materials • Methods of application • Evolution as asphalt
3. Describe logging roads	• “Corduroy” roads • Logging roads
4. Explain methods of placement	• Modern methods
5. Rationale for more roads	• WWII and Cold War • Movement of goods

Line (GAC): A ROAD BUILDING PRINCIPLES AND INTRODUCTION TO PAVING

Competency: A3 Describe paving technology

Objectives

To be competent in this area, the individual must be able to understand the process of paving technology from the plant to the train.

LEARNING TASKS	CONTENT
1. Describe two types of asphalt plants	• Drum • Batch
2. Describe asphalt ingredients	• Processed gravel • Asphalt cement • Oil and additives
3. Discuss required resources	• Quarries • Pits
4. Describe liquids used in asphalt production	• Group A / C – 85-100 Pen – typical PG 72-22 – altered oil
5. Describe asphalt recycling technology	• Conventional • Green Technologies
6. Describe the asphalt “Train”	• Hot in place

Line (GAC): A ROAD BUILDING PRINCIPLES AND INTRODUCTION TO PAVING

Competency: A4 Describe industry expectations of/for workers

Objectives

To be competent in this area, the individual must be able to:

- Understand the roles of each person in a paving team.
- Understand the professional expectations of this team.

LEARNING TASKS	CONTENT
1. Explain the importance of Health and Safety	• Content
2. Define types of paving jobs	• Raker • Screed • Asphalt Paver • Compaction Roller • Milling machine, etc.
3. Explain license requirements for paving equipment operators	• BC class 5 required to operate vehicles
4. Demonstrate professionalism	• Pride in the projects
5. Explain employer's responsibility to provide a safe environment	• To help maintain safe environment
6. Describe advancement opportunities	• Progression from labourer
7. Describe job security	• Job security based on high demand for skilled operators
8. Explain role of organized Labour	• Role of unions in the industry – collective bargaining process overview
9. Review working conditions	• Long hours (nights, weekends), seasonal work • Environment (outdoor/ weather conditions, dust/ fumes)